Examples:

1) **Asbestos:** These are fibrous and non-combustible silicates. Therefore they are used for thermal insulation material, brake linings, construction material and filters. Asbestos being carcinogenic silicates, their applications are restricted.

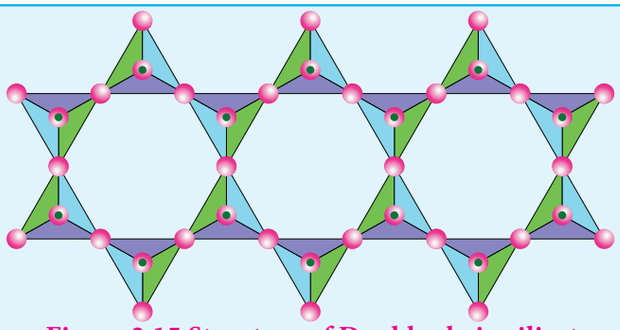


Figure 2.15 Structure of Double chain silicates

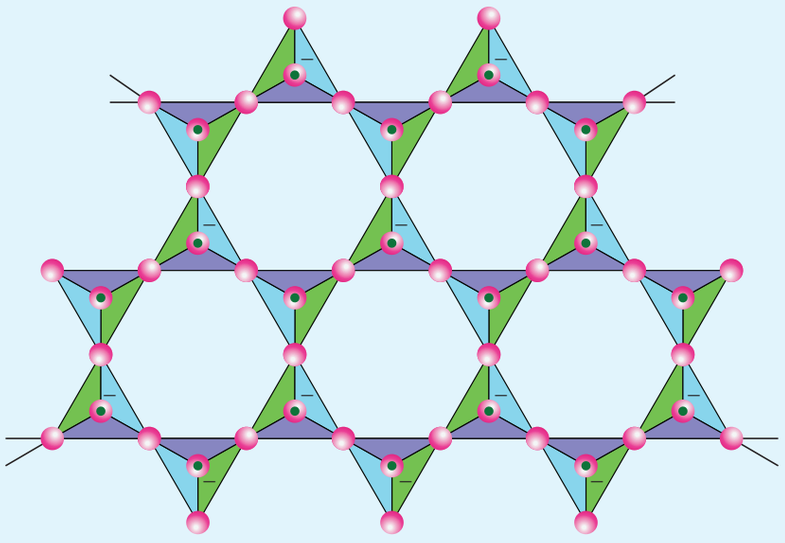


Figure 2.16 Structure of Sheet or phyllo silicates

Sheet or phyllo silicates

Silicates which contain $(Si_2O_5)_n^{2n-}$ are called sheet or phyllo silicates. In these, Each $[SiO_4]^{4-}$ tetrahedron unit shares three oxygen atoms with others and thus by forming two-dimensional sheets. These sheets silicates form layered structures in which silicate sheets are stacked over each other. The attractive forces between these layers are very weak, hence they can be cleaved easily just like graphite.

Example: Talc, Mica etc...

Three dimensional silicates (or tecto silicates):

Silicates in which all the oxygen atoms of $[SiO_4]^{4-}$ tetrahedra are shared with other tetrahedra to form three-dimensional network are called three dimensional or tecto silicates. They have general formula $(SiO_2)_n$.

Examples: Quartz

These tecto silicates can be converted into three dimensional aluminosilicates by replacing $[SiO_4]^{4-}$ units by $[AlO_4]^{5-}$ units. E.g. Feldspar, Zeolites etc.,

2.3.10 Zeolites:

Zeolites are three-dimensional crystalline solids containing aluminium, silicon, and oxygen in their regular three dimensional framework. They are hydrated sodium alumino silicates with general formula $Na_2O.(Al_2O_3).x(SiO_2).yH_2O$ (x=2 to 10; y=2 to 6).

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